



Student number: _____

2023

YEAR 12

Trial
EXAMINATION

Mathematics Extension 1

**General
Instructions**

- Reading time – 10 minutes
- Working time - 120 minutes
- Write using black or blue pen
- NESA approved calculators may be used
- A reference sheet is provided at the back of this paper

**Total marks:
70****Section I – 10 marks**

- Attempt Questions 1-10
- Allow about 15 minutes for this section
- Answer multiple choice questions by completely colouring in the appropriate circle on the answer sheet provided

Section II – 60 marks

- Attempt questions 11 – 16 in the answer booklet provided
- Allow about 1 hour and 45 minutes for this section
- Start each question on a new page showing all relevant mathematical reasoning and/or calculations

Section I

10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1-10 .

Question 1 (1 mark)

Which of the following is the remainder when the polynomial $P(x) = (x + 2)^3 + 2$ is divided by $(x - 2)$?

- A. 2
- B. 66
- C. $x - 2$
- D. $x + 2$

Question 2 (1 mark)

What is the value of $\lim_{x \rightarrow 0} \left(\frac{\sin \frac{1}{3}x}{2x} \right)$

- A. $\frac{1}{6}$
- B. $\frac{2}{3}$
- C. $\frac{3}{2}$
- D. 6

Question 3 (1 mark)

After t years the number N of individuals in a population is given by $N = 400 + 100e^{-0.1t}$. What is the difference between the initial population size and the limiting population size?

- A. 100
- B. 300
- C. 400
- D. 500

Question 4 (1 mark)

What is the acute angle between the vectors $\underline{\hat{i}} + 2\underline{\hat{j}}$ and $4\underline{\hat{i}} + 2\underline{\hat{j}}$ correct to the nearest degree?

- A. 18°
- B. 26°
- C. 32°
- D. 37°

Question 5 (1 mark)

A school committee consists of 8 members and a chairperson. The members are selected from 12 students. The chairperson is selected from 4 teachers. In how many ways could the committee be selected?

- A. ${}^{12}C_8 + {}^4C_1$
- B. ${}^{12}P_8 + {}^4P_1$
- C. ${}^{12}C_8 \times {}^4C_1$
- D. ${}^{12}P_8 \times {}^4C_1$

Question 6 (1 mark)

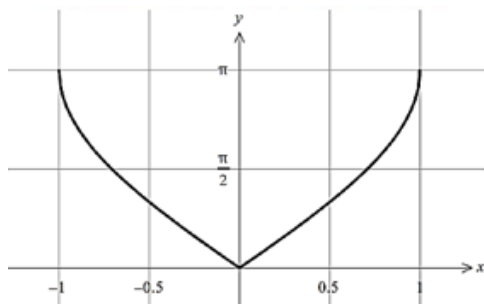
The vectors $\underline{p} = 4\underline{\hat{i}} + (a + 1)\underline{\hat{j}}$ and $\underline{q} = a\underline{\hat{i}} - 2\underline{\hat{j}}$ are perpendicular. What is the value of a ?

- A. -1
- B. 1
- C. $\frac{1}{3}$
- D. $-\frac{1}{3}$

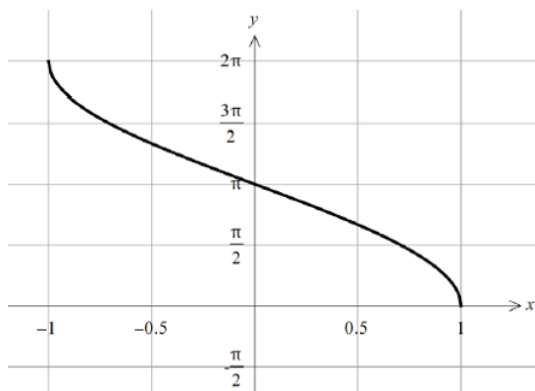
Question 7 (1 mark)

Which graph best represents $y = |2\cos^{-1}x - \pi|$?

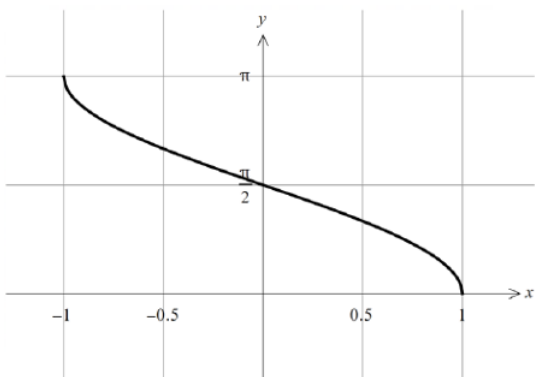
A.



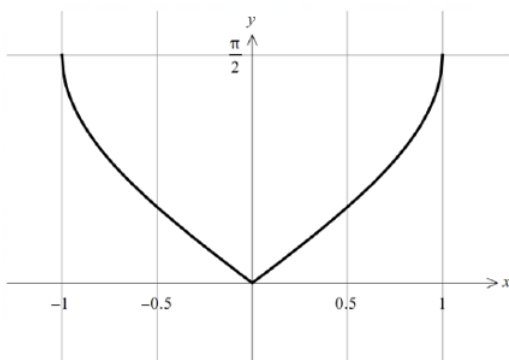
B.



C.



D.



Question 8 (1 mark)

A vertical tower AO of height h metres stands with its base A on horizontal ground. A stone is projected horizontally from the top O of the tower with speed $V \text{ ms}^{-1}$. The stone moves in a vertical plane under gravity where the acceleration due to gravity is $g \text{ ms}^{-2}$. At time t seconds its position vector relative to O is $\underline{r}(t) = (Vt)\underline{i} - \left(\frac{1}{2}gt^2\right)\underline{j}$. The stone hits the ground at an angle of 45° to the horizontal. What is the time of the impact?

- A. $\frac{V}{2g}$ seconds
- B. $\frac{V}{g}$ seconds
- C. $\frac{2V}{g}$ seconds
- D. $\frac{4V}{g}$ seconds

Question 9 (1 mark)

$g(x)$ is the inverse function of $f(x) = e^{x-1}$. Which one of these statements must be true for all x in the domain of $g(x)$?

- A. $g(x) > 0$
- B. $g(x) < 0$
- C. $g''(x) < 0$
- D. $g'(x) < 0$

Question 10 (1 mark)

$$\sin(3x + x) - \sin(3x - x) =$$

- A. $-2 \sin 3x \sin x$
- B. $2 \cos 3x \sin x$
- C. $2 \cos 3x \cos x$
- D. $2 \sin 3x \sin x$

Section II

60 marks

Attempt Questions 11-16

Allow about 1 hour and 45 minutes for this section

Answer each question on a separate page in the writing booklet.

Extra writing booklets are available

For questions in Section II, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (10 marks)

- (a) Solve the inequality $\frac{2x-1}{x+2} > 1$ [3]
- (b) Use the substitution $u = 6 - x$ to find the exact value of $\int_1^6 x\sqrt{6-x} \, dx$ [3]
- (c) A group of 40 new Year 7 students is going to be randomly divided into two classes A and B of 20 students each. Four of the students, Anne, Ben, Carrie and David have been together in the same class since they started school and are close friends.
- i. What is the probability that the four friends will be in the same class? [2]
- ii. What is the probability that exactly three of the friends will be in the same class? [2]

Question 12 (10 marks)

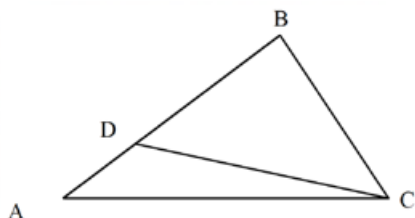
- (a) Find the exact value of $\int_{\sqrt{2}}^{\sqrt{3}} \frac{1}{\sqrt{4-x^2}} \, dx$ [3]
- (b) Find the cartesian equation for the curve with parametric equations $x = 1 + 2 \cos 2t$ and $y = 2 + 2 \sin 2t$ [2]
- (c) Prove by principle of mathematical induction that $7^n - 3^n$ is divisible by 4 for $n \geq 1$. [3]
- (d) Prove the identity $\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$ [2]

Question 13 (11 marks)

- (a) If $t = \tan \frac{x}{2}$, show that $\frac{1 + \cos x + \sin x}{1 - \cos x + \sin x} = \cot \frac{x}{2}$. [3]
- (b) Using t - formulae solve the equation $\sin \theta + \cos \theta = \frac{1}{2}$ for $[0, 2\pi]$. Answer in radians to 3 decimal places. [4]
- (c) i. Express $\cos x - \sqrt{3} \sin x$ in the form $R \cos(x + \alpha)$, where α is an acute angle. [2]
- ii. Hence, solve $\cos x - \sqrt{3} \sin x = 1$ for $[-\pi, \pi]$ [2]

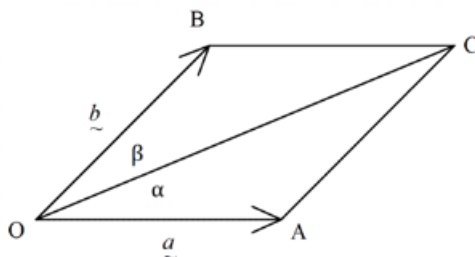
Question 14 (10 marks)

- (a) If α, β and γ are roots of $x^3 - 5x^2 + 7x + 5 = 0$
- i. Find $\alpha + \beta + \gamma$ [1]
- ii. Find $\alpha\beta + \beta\gamma + \gamma\alpha$ [1]
- (b) Given $\underline{u} = 2\underline{i} + 3\underline{j}$ and $\underline{v} = -2\underline{i} + 4\underline{j}$. Find $\text{proj}_{\underline{u}} \underline{v}$ [2]
- (c) In $\triangle ABC$, D is a point on AB , where $|\overrightarrow{AD}| : |\overrightarrow{DB}| = 2:3$ [2]



Given $\overrightarrow{AD} = \underline{a}$, $\overrightarrow{AC} = \underline{b}$ and $\overrightarrow{CB} = \underline{c}$. Show that $\underline{b} = \frac{1}{2}(5\underline{a} - 2\underline{c})$

- (d) $OACB$ is rhombus. $\overrightarrow{OA} = \underline{a}$, $\overrightarrow{OB} = \underline{b}$, $\angle AOC = \alpha$ and $\angle COB = \beta$.



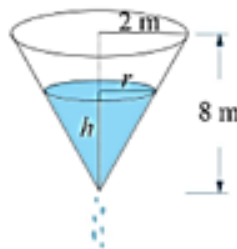
- i. For the vectors in the diagram above prove that $\underline{a} \cdot (\underline{a} + \underline{b}) = \underline{b} \cdot (\underline{a} + \underline{b})$ [2]
- ii. Hence, prove diagonal OC bisects $\angle AOB$. [2]

Question 15 (10 marks)

- (a) Jerry stands at the top of a building 30m tall and throws a ball with a velocity of 15ms^{-1} at an angle of 45° above the horizontal. The ball eventually reaches the ground.
- Derive the vector displacement of the ball in the form $\underline{s} = x\underline{i} + y\underline{j}$. You may assume that $g = 10\text{ms}^{-2}$. [3]
 - Find the time taken for the ball to reach the ground. [1]
- (b) Consider the function $f(x) = \sin^{-1}(x - 1)$.
- Find the domain of the function. [1]
 - Sketch the graph of the curve $y = f(x)$ showing the endpoints and the x -intercept(s). [2]
 - The region in the first quadrant bounded by the curve $y = f(x)$ and the y -axis between the lines $y = 0$ and $y = \frac{\pi}{2}$ is rotated through one complete revolution about the y -axis. Find in simplest exact form the volume of the solid of revolution. [3]

Question 16 (9 marks)

- (a) Find the constant term in the expression $\left(x + \frac{2}{x}\right)^6$ [3]
- (b) An inverted conical container is 8 metres deep and has a base radius of 2 metres. Water is leaking from the container at a constant rate of $\frac{dV}{dt} = 0.1$ metres/hour, where V is the volume of the water in the container. Assume the container is full initially.



- Show that $V = \frac{\pi}{48}h^3$, where h is the height of the remaining water in the container. [1]
- Hence, find the height of water in the container when $\frac{dh}{dt} = 0.02$ metres/hour, correct to two decimal places. [2]

Question 16 continued on next page...

Question 16 (continued)

- (c) Newton's Law of cooling states that when an object at temperature $T^\circ\text{C}$ is placed in an environment at temperature T_0° , the rate of temperature loss is given by the equation

$$\frac{dT}{dt} = k(T - T_0)$$

Where t is the time in seconds and k is a constant.

A packet of peas, initially at 24°C , is placed in a snap freeze refrigerator in which the internal temperature is maintained at -40°C .

- i. Show that $T = -40 + 64e^{kt}$ satisfies the cooling equation $\frac{dT}{dt} = k(T - T_0)$. [1]
- ii. After 5 seconds, the temperature of the packet is 19°C . How long will it take for the temperature of the packet to reduce to 0°C ? [2]

End of paper

①

$$① \quad p(x) = (x+2)^3 + 2$$

$$\begin{aligned} \text{Rem} &= p(2) = (2+2)^3 + 2 \\ &= 66 \end{aligned}$$

Ans. B.

$$② \quad \lim_{x \rightarrow 0} \left(\frac{\sin\left(\frac{1}{3}x\right)}{2x} \right)$$

$$= \frac{1}{6} \lim_{x \rightarrow 0} \left(\frac{\sin \frac{1}{3}x}{\frac{1}{3}x} \right)$$

$$= \frac{1}{6} \times 1$$

$$= \frac{1}{6}$$

Ans. A

$$③ \quad N = 400 + 100e^{-0.1t}$$

when $t = 0$

$$N = 500$$

$$\text{and } \lim_{t \rightarrow \infty} N = 400$$

Ans. A

Hence, initial population - limiting population

$$= 500 - 400$$

$$= 100$$

(4) Acute angle between
 $\underline{\underline{i}} + 2\underline{\underline{j}}$ and $4\underline{\underline{i}} + 2\underline{\underline{j}}$

Ans. D

(2)

$$\text{let } \underline{u} = \underline{\underline{i}} + 2\underline{\underline{j}} ; \underline{v} = 4\underline{\underline{i}} + 2\underline{\underline{j}}$$

$$\cos \theta = \frac{\underline{u} \cdot \underline{v}}{|\underline{u}| |\underline{v}|}$$

$$\cos \theta = \frac{4 + 4}{\sqrt{5} \times \sqrt{20}}$$

$$\cos \theta = \frac{8}{5 \times 2}$$

$$\cos \theta = \frac{4}{5}$$

$$\theta \simeq 36^\circ 52'$$

$$\theta \simeq 37^\circ$$

(5) ${}^{12}C_8 \times {}^4C_4$

Ans. C

(6) $\begin{pmatrix} 4 \\ a+1 \end{pmatrix} \cdot \begin{pmatrix} a \\ -2 \end{pmatrix} = 0$

Ans. B

$$4a - 2(a+1) = 0$$

$$2a - 2 = 0$$

$$a = 1$$

Ans. A

(8) Velocity vector is
 45° below horizontal
 when $\frac{\dot{y}}{\dot{x}} = -1$

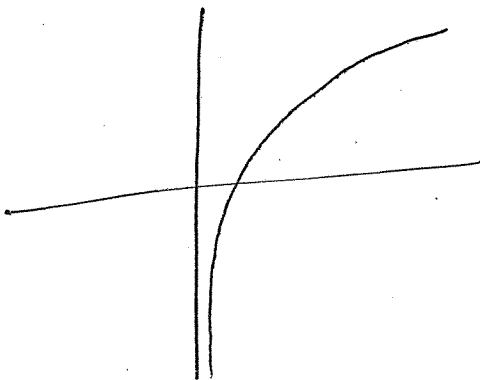
Ans. B

$$\therefore \frac{-gt}{v} = -1$$

$$\therefore t = \frac{v}{g} \text{ seconds}$$

(9) $g''(x) < 0$
 Graph of $g(x)$

Ans. C



(10) $\sin(3x+a) - \sin(3x-a)$
 $2 \cos 3x \sin a$

Ans. B

Question 11

$$(a) \frac{2x-1}{x+2} > 1$$

$$\frac{(2x-1)(x+2)^2}{(x+2)} > (x+2)^2 \text{ and } x \neq -2$$

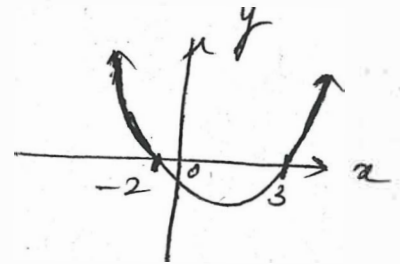
$$(2x-1)(x+2) > (x+2)^2$$

$$(2x-1)(x+2) - (x+2)^2 > 0$$

$$(x+2) \{ (2x-1) - (x+2) \} > 0$$

$$(x+2)(x-3) > 0$$

$$\therefore x < -2 \text{ or } x > 3$$



(6) let $u = 6 - x$ Evaluate $\int_1^6 x \sqrt{6-x} \, dx$

$$\int_1^6 x \sqrt{6-x} \, dx$$

$$= - \int_5^0 (6-u) \sqrt{u} \, du$$

$$= \int_0^5 (6-u) \sqrt{u} \, du$$

$$= \int_0^5 (6u^{1/2} - u^{3/2}) \, du$$

$$= \left[\frac{6 \times 2}{3} u^{3/2} - \frac{2}{5} u^{5/2} \right]_0^5$$

$$= \left[4 u^{3/2} - \frac{2}{5} u^{5/2} \right]_0^5$$

$$= 4 \times 5 \sqrt{5} - \frac{2}{5} \times 5^2 \times \sqrt{5}$$

$$= \underline{\underline{10\sqrt{5}}}$$

$$u = 6 - x$$

$$du = -dx$$

limits:

$$x=1 \Rightarrow u=5$$

$$x=6 \Rightarrow u=0$$

Question 11 - continued

- © i, put 4 friends in class A, then need to choose 16 more students from 36

$$\text{Probability all 4 friends in class A} = \frac{{}^{36}C_{16}}{{}^{40}C_{20}}$$

could also be in class B

$$\text{Probability all in same class} = 2 \times \frac{{}^{36}C_{16}}{{}^{40}C_{20}}$$

3 d.p.)

- ii) put 3 friends in class A
These 3 friends can be chosen in 4C_3 ways
then we need to choose 17 more students from 36 (because 4th friend can't be in class A)

$$\text{Probability exactly 3 friends in class A} = \frac{{}^4C_3 \times {}^{36}C_{17}}{{}^{40}C_{20}}$$

could also be in class B

$$\begin{aligned} \text{Probability exactly 3 friends in same class} &= 2 \times \frac{{}^4C_3 \times {}^{36}C_{17}}{{}^{40}C_{20}} \\ &\approx 0.499 \text{ (3 d.p.)} \end{aligned}$$

Question 12

(7)

$$(a) \int_{\sqrt{2}}^{\sqrt{3}} \frac{1}{\sqrt{4-x^2}} dx$$

$$= \left[\sin^{-1}\left(\frac{x}{2}\right) \right]_{\sqrt{2}}^{\sqrt{3}}$$

$$= \sin^{-1}\left(\frac{\sqrt{3}}{2}\right) - \sin^{-1}\left(\frac{1}{2}\right)$$

$$= \frac{\pi}{3} - \frac{\pi}{6}$$

$$= \frac{\pi}{6}$$

Question 12 - continued

⑧

(b) $x = 1 + 2 \cos 2t$; $y = 2 + 2 \sin 2t$

$$x - 1 = 2 \cos 2t$$

$\hookrightarrow$ ①

$$y - 2 = 2 \sin 2t$$

$\hookrightarrow$ ②

squaring and adding eq ① & ②

$$(x-1)^2 + (y-2)^2 = 4 \cos^2 2t + 4 \sin^2 2t$$

$$(x-1)^2 + (y-2)^2 = 4 (\sin^2 2t + \cos^2 2t)$$

$\therefore (x-1)^2 + (y-2)^2 = 4$ is the Cartesian equation

Question 12 continued

(c) let $p(n)$ be the proposition that $7^n - 3^n$ is divisible by 4 for $n \geq 1$

$$\rightarrow \text{when } n=1; \text{ LHS} = 7^1 - 3^1 = 4 \\ = 4 \times 1$$

$\therefore p(n)$ is true when $n=1$

$\rightarrow$ Assume that $p(n)$ is true when $n=k$
i.e., assume that $7^k - 3^k = 4M$ where $M \in \mathbb{Z}^+$
 $\hookrightarrow \text{eq (1)}$

$\rightarrow$ Required to prove that $p(n)$ is true when $n=k+1$
i.e. prove that $7^{k+1} - 3^{k+1} = 4P$, where $P \in \mathbb{Z}^+$

$$\text{L.H.S} = 7^{k+1} - 3^{k+1}$$

$$= 7 \cdot 7^k - 3 \cdot 3^k$$

$$= 7(4M + 3^k) - 3 \cdot 3^k \quad (\text{from eq (1), by assumption})$$

$$= 7 \times 4M + 7 \cdot 3^k - 3 \cdot 3^k$$

$$= 7 \times 4M + 3^k(7-3)$$

$$= 7 \times 4M + 4 \cdot 3^k$$

$$= 4(7M + 3^k)$$

$$= 4P, \text{ where } P \in \mathbb{Z}^+$$

$\therefore$ By principle of Mathematical induction if $p(k)$ is true
th $p(k+1)$ is true

Question 12 - continued

(d) Prove $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$

$$\text{L.H.S} = \cos 3\theta$$

$$= \cos (2\theta + \theta)$$

$$= \cos 2\theta \cos \theta - \sin 2\theta \sin \theta$$

$$= (2\cos^2\theta - 1)\cos\theta - (2\sin\theta\cos\theta)\sin\theta$$

$$= 2\cos^3\theta - \cos\theta - 2\sin^2\theta\cos\theta$$

$$= 2\cos^3\theta - \cos\theta - 2\cos\theta(1 - \cos^2\theta)$$

$$= 2\cos^3\theta - \cos\theta - 2\cos\theta + 2\cos^3\theta$$

$$= 4\cos^3\theta - 3\cos\theta$$

$$= \text{R.H.S}, \text{ as required}$$

13

(a) If $t = \tan \frac{\theta}{2}$

∴ Show that $\frac{1 + \cos \alpha + \sin \alpha}{1 - \cos \alpha + \sin \alpha} = \cot \frac{\alpha}{2}$

$$1 + \cos \alpha + \sin \alpha = 1 + \frac{1-t^2}{1+t^2} + \frac{2t}{1+t^2}$$

$$= \frac{(1+t^2) + (1-t^2) + 2t}{1+t^2}$$

$$= \frac{2(1+t)}{1+t^2}$$

$$1 - \cos \alpha + \sin \alpha = 1 - \frac{1-t^2}{1+t^2} + \frac{2t}{1+t^2}$$

$$= \frac{(1+t^2) - (1-t^2) + 2t}{1+t^2}$$

$$= \frac{2t(1+t)}{1+t^2}$$

Hence $\frac{1 + \cos \alpha + \sin \alpha}{1 - \cos \alpha + \sin \alpha} = \frac{1}{t} = \cot \frac{\alpha}{2}$

Question 15 continued

(a) (ii)

Solve $\sin \theta + \cos \theta = \frac{1}{2}$ for $[0, 2\pi]$

(12)

$$\sin \theta + \cos \theta = \frac{1}{2}$$

$$\frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2} = \frac{1}{2}$$

$$\frac{1+2t-t^2}{1+t^2} = \frac{1}{2}$$

$$2(1+2t-t^2) = 1+t^2$$

$$2+4t-2t^2 = 1+t^2$$

$$3t^2 - 4t - 1 = 0$$

$$t = \frac{4 \pm \sqrt{16 + 4(3)(1)}}{6}$$

$$t = \frac{4 \pm \sqrt{28}}{6} = \frac{2 \pm 2\sqrt{7}}{3}$$

$$\tan \frac{\theta}{2} = \frac{2 \pm \sqrt{7}}{3}, \quad 0 \leq \theta \leq 2\pi; \quad 0 \leq \frac{\theta}{2} \leq \pi$$

$$\frac{\theta}{2} = \tan^{-1}\left(\frac{2 \pm \sqrt{7}}{3}\right) = 0.997, \quad \pi - 0.212$$

$$= 0.997, \quad 2.930 \text{ radians}$$

Test ~~off~~ for $\theta = \pi$ which is not a solution.

$$\therefore \theta = 1.995, \quad 5.859 \text{ radians}$$

$$b.i, \cos x - \sqrt{3} \sin x = R \cos(x + \alpha)$$

$$\cos x - \sqrt{3} \sin x = R \cos x \cos \alpha - R \sin x \sin \alpha$$

Now comparing L.H.S & R.H.S

we can see that

$$R \cos \alpha = 1 \quad \text{and} \quad R \sin \alpha = \sqrt{3}$$

$$\hookrightarrow \textcircled{1}$$

$$\hookrightarrow \textcircled{2}$$

Squaring and adding eq $\textcircled{1}$ & $\textcircled{2}$

$$R^2 \cos^2 \alpha + R^2 \sin^2 \alpha = 1 + 3$$

$$R^2 (\sin^2 \alpha + \cos^2 \alpha) = 4$$

$$R^2 = 4 \Rightarrow R = 2 \text{ as } R > 0$$

$$\text{Since } \cos \alpha = \frac{1}{2} > 0 \text{ and } \sin \alpha = \frac{\sqrt{3}}{2} > 0$$

α is in the 1st Quadrant and $\alpha = \frac{\pi}{3}$

$$\therefore \cos x - \sqrt{3} \sin x = R \cos \left(x + \frac{\pi}{3}\right)$$

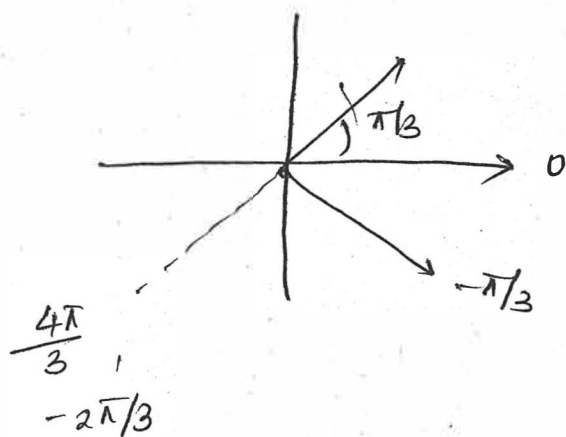
Question 13 - Continued

14

b, ii, $\cos x - \sqrt{3} \sin x = 1$ for $[-\pi, \pi]$

$$\therefore 2 \cos \left(x + \frac{\pi}{3} \right) = 1$$

$$\therefore \cos \left(x + \frac{\pi}{3} \right) = \frac{1}{2}$$



Changing the domain

$$-\pi \leq x \leq \pi$$

$$\left(-\pi + \frac{\pi}{3} \right) \leq \left(x + \frac{\pi}{3} \right) \leq \left(\pi + \frac{\pi}{3} \right)$$

$$-\frac{2\pi}{3} \leq \left(x + \frac{\pi}{3} \right) \leq \frac{4\pi}{3}$$

$\cos \left(x + \frac{\pi}{3} \right) = \frac{1}{2}$; related acute angle = $\frac{\pi}{3}$

$$\therefore x + \frac{\pi}{3} = -\frac{\pi}{3}, \frac{\pi}{3}$$

$$\therefore x = -\frac{2\pi}{3}, 0$$

Question 14

(15)

(a) α, β and γ are roots of $x^3 - 5x^2 + 7x + 5 = 0$

$$\begin{aligned} \text{i, } \alpha + \beta + \gamma &= -\frac{b}{a} \\ &= -(-5/1) \\ &= 5 \end{aligned}$$

$$\begin{aligned} \text{ii, } \alpha\beta + \beta\gamma + \gamma\alpha &= \frac{c}{a} \\ &= \frac{7}{1} \\ &= 7 \end{aligned}$$

(b) $\underline{u} = 2\underline{i} + 3\underline{j}$ and $\underline{v} = -2\underline{i} + 4\underline{j}$

$$\begin{aligned} \text{Proj}_{\underline{u}} \underline{v} &= \frac{\underline{u} \cdot \underline{v}}{|\underline{u}|^2} \cdot \underline{u} \\ &= \frac{2(-2) + (3 \times 4)}{2^2 + 3^2} \cdot \begin{pmatrix} 2 \\ 3 \end{pmatrix} \\ &= \frac{8}{13} \left(\frac{2}{3} \right) \text{ or } \frac{8}{13} (2\underline{i} + 3\underline{j}) \end{aligned}$$

Question 14 - Continued

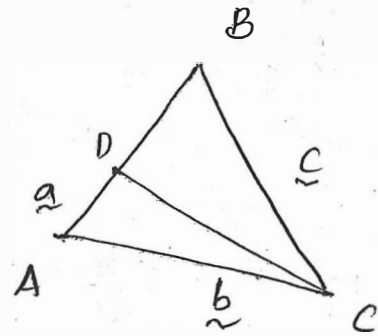
16

c) Given $|\vec{AD}| : |\vec{DB}| = 2:3$

$$\vec{AB} = \underline{a}$$

$$\vec{AC} = \underline{b}$$

$$\vec{CB} = \underline{c}$$



Show that $\underline{b} = \frac{1}{2} (5\underline{a} - 2\underline{c})$

$$|\vec{DB}| : |\vec{AB}| = 3:2 \text{ and } \vec{AB} = \underline{a}$$

$$\vec{DB} = \frac{3}{2} \underline{a}$$

$$\vec{AC} = \vec{AB} + \vec{BC}$$

$$\underline{b} = \vec{AD} + \vec{DB} - \vec{CB}$$

$$= \underline{a} + \frac{3}{2} \underline{a} - \underline{c}$$

$$= \frac{5}{2} \underline{a} - \underline{c}$$

$$= \frac{1}{2} (5\underline{a} - 2\underline{c})$$

(d) i, Prove that $\underline{a} \cdot (\underline{a} + \underline{b}) = \underline{b} \cdot (\underline{a} + \underline{b})$

$$\text{L.H.S} = \underline{a} \cdot (\underline{a} + \underline{b})$$

$$= \underline{a} \cdot \underline{a} + \underline{a} \cdot \underline{b}$$

$$= |\underline{a}|^2 + \underline{a} \cdot \underline{b}$$

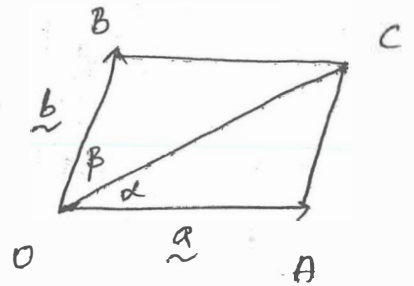
$$= |\underline{b}|^2 + \underline{a} \cdot \underline{b} \quad (\text{Since } OACB \text{ is a rhombus})$$

$$= \underline{b} \cdot \underline{b} + \underline{a} \cdot \underline{b}$$

$$= \underline{b} (\underline{b} + \underline{a})$$

$$= \underline{b} (\underline{a} + \underline{b})$$

$$= \text{R.H.S}$$



(d) ii Prove diagonal OC bisects $\angle AOB$.

from i, $\underline{a} \cdot (\underline{a} + \underline{b}) = \underline{b} \cdot (\underline{a} + \underline{b})$

$$|\underline{a}| \cdot |\underline{a} + \underline{b}| \cos \alpha = |\underline{b}| \times |\underline{a} + \underline{b}| \cos \beta$$

$$|\underline{a}| |\underline{a} + \underline{b}| \cos \alpha = |\underline{a}| \times |\underline{a} + \underline{b}| \cos \beta$$

$$\cos \alpha = \cos \beta$$

$$(\text{Since } |\underline{a}| = |\underline{b}|)$$

$$\alpha = \beta$$

$\therefore$ OC bisects $\angle AOB$

Question 15

$$(a) \dot{y} = -g \quad \ddot{x} = 0$$

$$\dot{y} = \int -g dt \quad \dot{x} = C_2$$

Velocity: $\dot{y} = -gt + C_1$

when $t=0$, $\dot{y} = \frac{15\sqrt{2}}{2}$

$$\therefore \frac{15\sqrt{2}}{2} = -g(0) + C_1$$

$$\therefore C_1 = \frac{15\sqrt{2}}{2}$$

and $\dot{y} = -gt + \frac{15\sqrt{2}}{2}$

when $t=0$ $\dot{x} = \frac{15\sqrt{2}}{2}$

$$\therefore C_2 = \frac{15\sqrt{2}}{2} \text{ and}$$

$$\dot{x} = \frac{15\sqrt{2}}{2}$$

Displacement:

$$y = \int (-gt + \frac{15\sqrt{2}}{2}) dt$$

$$y = \frac{-gt^2}{2} + \frac{15\sqrt{2}}{2} t + C_3$$

when $t=0$; $y=30$

$$\therefore 30 = -0 + 0 + C_3$$

$$\therefore C_3 = 30 \text{ and}$$

$$y = \frac{-gt^2}{2} + \frac{15\sqrt{2}}{2} t + 30$$

using $g=10$

$$\therefore \underline{s} = \left(\frac{15\sqrt{2}}{2} t \right) \underline{i} + \left(-5t^2 + \frac{15\sqrt{2}}{2} t + 30 \right) \underline{j}$$

$$x = \int \frac{15\sqrt{2}}{2} dt$$

$$x = \frac{15\sqrt{2}}{2} t + C_4$$

when $t=0$, $x=0$

$$\therefore C_4 = 0 \text{ and}$$

$$x = \frac{15\sqrt{2}}{2} t$$

(19)

Question 15 - continued

aii, For the ball to reach the ground the component ⁽²⁰⁾ of $\vec{v}$ must equal zero.

$$-5t^2 + \frac{15\sqrt{2}}{2}t + 30 = 0$$

$$t = \frac{-\frac{15\sqrt{2}}{2}}{2(-5)} \pm \frac{\sqrt{\left(\frac{15\sqrt{2}}{2}\right)^2 - 4(-5)(30)}}{2(-5)}$$

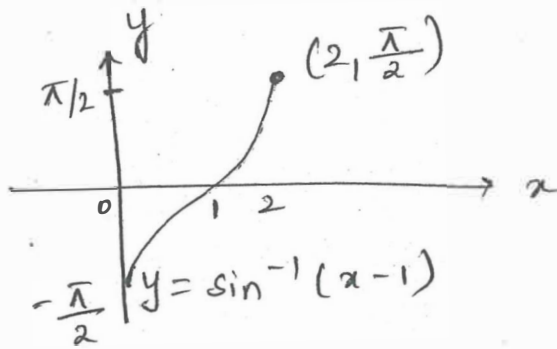
$$= \frac{-\frac{15\sqrt{2}}{2}}{-10} \pm \sqrt{712.5}$$

$$\approx 3.7 \text{ seconds}$$

(b) i, $f(x) = \sin^{-1}(x-1)$

Domain : $-1 \leq (x-1) \leq 1$

$0 \leq x \leq 2$



iii, $V = \pi \int_0^{\pi/2} (1 + \sin y)^2 dy$

$= \pi \int_0^{\pi/2} (1 + 2\sin y + \sin^2 y) dy$

$= \pi \int_0^{\pi/2} (1 + 2\sin y + \frac{1}{2}(1 - \cos 2y)) dy$

$= \pi \left[y - 2\cos y + \frac{1}{2}y - \frac{1}{4}\sin 2y \right]_0^{\pi/2}$

$= \pi \left[\frac{3}{2}y - 2\cos y - \frac{1}{4}\sin 2y \right]_0^{\pi/2}$

$= \pi \left[\left(\frac{3\pi}{4} - 0 - 0 \right) - (0 - 2 - 0) \right]$

$= \pi \left(\frac{3\pi}{4} + 2 \right) \text{ Cubic units}$

(a) constant term in $\left(x + \frac{2}{x}\right)^6$

$$\begin{aligned} T_{k+1} &= {}^6C_k x^{6-k} \left(\frac{2}{x}\right)^k \\ &= {}^6C_k \times 2^k x^{6-k} \left(\frac{1}{x^k}\right) \\ &= {}^6C_k \times 2^k x^{6-2k} \end{aligned}$$

for the constant term: $6-2k=0$

$$\therefore k=3$$

Hence, the constant term is ${}^6C_3 \times 2^3 = 160$

(b) i, Using similar triangles

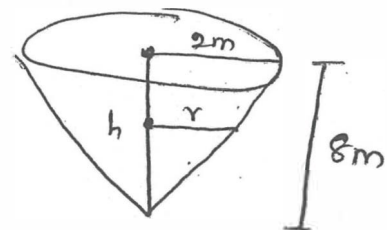
$$\frac{2}{r} = \frac{8}{h}$$

$$\therefore r = \frac{h}{4}$$

Now $V = \frac{1}{3} \pi r^2 h$

$$V = \frac{1}{3} \pi \left(\frac{h}{4}\right)^2 \cdot h$$

$$V = \frac{\pi}{48} h^3$$



(b) ii, $V = \frac{\pi}{48} h^3$

(23)

$$\frac{dv}{dt} = \frac{dv}{dh} \times \frac{dh}{dt}$$

$$0.1 = \frac{d}{dh} \left(\frac{\pi}{48} h^3 \right) \times 0.02$$

$$\frac{0.1}{0.02} = 3 \times \frac{\pi}{48} h^2$$

$$5 = \frac{\pi}{16} h^2$$

$$h^2 = \frac{80}{\pi}$$

$$\therefore h \simeq 5.05 \text{ m}$$

Ans

C. i, Substituting $T = -40 + 64e^{kt}$ in $\frac{dT}{dt} = K(T - T_0)$

$$\text{L.H.S.} : \frac{dT}{dt}$$

$$= \frac{d}{dt} (-40 + 64e^{kt})$$

$$= 0 + 64Ke^{kt}$$

$$= K(64e^{kt})$$

$$= K(T + 40)$$

$$= K(T - (-40))$$

$$= K(T - T_0), T_0 = -40$$

$$= \text{R.H.S.}$$

when $t=5$, $T=19$

(25)

$$19 = -40 + 64e^{5k}$$

$$e^{5k} = \frac{59}{64}$$

$$5k = \ln\left(\frac{59}{64}\right)$$

$$k = \frac{1}{5} \ln\left(\frac{59}{64}\right)$$

Sub $T=0$

$$0 = -40 + 64e^{kt}$$

$$40 = 64e^{kt}$$

$$e^{kt} = \frac{40}{64}$$

$$kt = \ln\left(\frac{40}{64}\right)$$

$$t = \frac{1}{k} \left(\ln\left(\frac{40}{64}\right) \right)$$

$$t = \frac{\ln(40/64)}{(\frac{1}{5}) \ln(59/64)}$$

$t \approx 29$ seconds -